Al Liu

Rust Systems Engineer & Open-Source Maintainer

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PROFILE

Rust systems engineer and open-source maintainer specializing in performance-sensitive infrastructure, asynchronous networking, distributed protocols, storage engines, and concurrency. My experience spans production financial systems, decentralized networking, blockchain infrastructure, and reusable developer libraries. I work across protocol implementation, runtime abstractions, concurrent data processing, and the practical work of making systems reliable.

TECHNICAL EXPERTISE

Languages	Rust / Go / SQL
Systems & networking	Linux / TCP / UDP / QUIC / WebSocket / Async I/O / Protobuf
Distributed systems	SWIM / Gossip / Service discovery / Consensus / Fault tolerance
Storage & performance	MVCC / Concurrent data structures / Skiplist / Caching / Zero-copy / Memory-mapped I/O
Additional technologies	WebAssembly / WASI / GraphQL

PROFESSIONAL EXPERIENCE

Confidential Financial Technology Company

Jan 2026 - Present

Senior Rust Systems Engineer

Full-time / Remote

Build performance-sensitive Rust infrastructure for real-time financial systems, with a focus on reliability.

- Redesigned the concurrency model around a thread-per-core, shared-nothing architecture with pre-allocated resources and mutex-free busy-loop execution.
- Reduced a critical service's p99.9 tail latency from several hundred milliseconds to 1-2 ms, p99 from tens of milliseconds to below 1 ms, and p90 from 100+ microseconds to 10-20 microseconds.
- Develop and maintain asynchronous networking and concurrent data pipelines for real-time financial systems.
- Improve production reliability and resource efficiency across the Rust infrastructure.

Freenet

May 2024 - Jul 2024

Rust Engineer

Part-time / Remote

Contributed to Freenet’s decentralized networking infrastructure through work in freenet-core and freenet-stdlib, covering node setup, network configuration, and operational reliability.

- Contributed to decentralized networking infrastructure in freenet-core and freenet-stdlib.
- Worked on node initialization and configuration in the Rust codebase.
- Contributed to hostname resolution and transport-key handling.
- Improved event-log reliability and the maintainability of the project workspace.

Status

Nov 2022 - Mar 2024

Rust Software Engineer - Nomos

Full-time / Remote

Worked on Nomos Layer 1 blockchain infrastructure, with contributions spanning consensus simulation and privacy-preserving communication between nodes.

- Developed simulation tooling for the Carnot distributed consensus algorithm, supporting the study of protocol behavior.
- Contributed Rust infrastructure supporting the Nomos Layer 1 blockchain.
- Worked on privacy-preserving networking, including implementation work on mixnets.

Polar Sync

Jan 2022 - Sep 2023

Rust and Go Engineer

Full-time contract / Remote

Worked across Rust and Go engineering, with a focus on code-generation infrastructure and the abstractions connecting application schemas, queries, and database functionality.

- Developed Rust procedural macros and reusable code-generation infrastructure.
- Worked on GraphQL and SQL functionality, including integration with The Graph subgraphs.
- Designed reusable abstractions for database and query integration across these tools.

OPEN-SOURCE ENGINEERING

Maintain a collection of Rust infrastructure libraries covering language tooling, distributed membership and discovery, asynchronous I/O, file operations, caching, and concurrent data structures. The public source provides concrete examples of protocol design, library APIs, and systems-level implementation.

Use Sans-I/O boundaries, runtime-independent abstractions, validated types, and explicit ownership to separate core logic from its environment. Public work includes SWIM and Lifeguard membership, GraphQL tooling, parser combinators, MVCC storage, memory mapping, and synchronization primitives.

Explore the complete project collection

EDUCATION

University of Auckland

Bachelor of Science with Honours in Computer Science

2019 - 2022 / Graduated November 2022

Completed the degree while also working full-time as a Rust and Go engineer during 2022.

2022 University of Auckland Sub-Doctoral Postgraduate Accommodation Scholarship